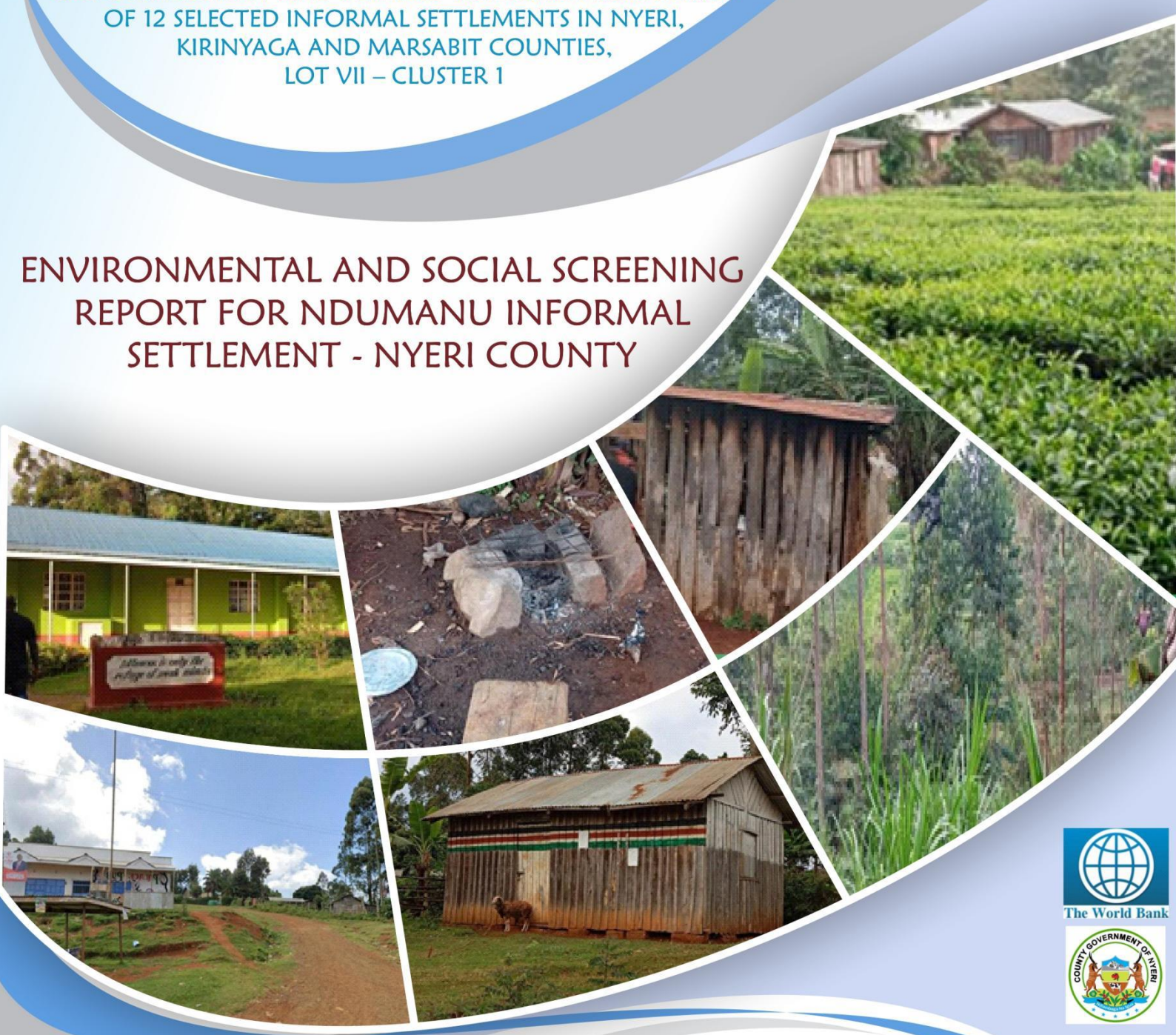
KENYA VISION 2030
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MINISTRY OF LANDS, PUBLIC WORKS, HOUSING & URBAN DEVELOPMENT
STATE DEPARTMENT FOR HOUSING AND URBAN DEVELOPMENT

SECOND KENYA INFORMAL SETTLEMENTS IMPROVEMENT PROJECT (KISIP 2)

CONSULTANCY SERVICES TO UNDERTAKE PHYSICAL PLANNING, CADASTRAL SURVEY AND DETAILED TOPOGRAPHICAL SURVEY OF 12 SELECTED INFORMAL SETTLEMENTS IN NYERI, KIRINYAGA AND MARSABIT COUNTIES, LOT VII – CLUSTER 1

ENVIRONMENTAL AND SOCIAL SCREENING REPORT FOR NDUMANU INFORMAL SETTLEMENT - NYERI COUNTY



The World Bank

COUNTY GOVERNMENT OF NYERI

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EXECUTIVE SUMMARY

The Government of Kenya, in collaboration with the World Bank (WB), initiated the Second Kenya Informal Settlements Improvements Project (KISIP2) to build upon the successes achieved through KISIP1 and to extend the advantages of the project to a larger population within informal settlements. In contrast to KISIP1, the new project incorporates innovative strategies and initiatives aimed at amplifying its positive effects on the communities involved. The project development objective is to improve access to basic services and tenure security of residents in participating urban informal settlements and strengthen institutional capacity for slum upgrading in Kenya. This will be accomplished through investments in infrastructure, guided by plans created in collaboration with the resident community; facilitating physical planning, land surveying, and the issuance of land ownership documents for individuals living in informal settlements; and by enhancing the capabilities of County administrations to fulfill their responsibilities effectively.

An Environmental and Social Screening Report serves as a tool to assess the potential impacts that a proposed project could impose on the environment and surrounding communities. It identifies potential risks and establishes the extent of additional environmental and social assessments needed, depending on the expected impacts of the project. This process facilitates the development of necessary mitigation strategies.

This report presents the findings of environmental and social screening conducted in Ndumanu informal settlement in Nyeri county. The settlement is located off the Itundu-Ragati Road in Magutu ward, Mathira East sub-county of Nyeri County, Kenya. The ward borders Ruguru Ward, Karatina Town Ward, Iriani Ward and Kirinyaga County. It measures approximately 8.5 Hectares.

The screening methodology applied comprised of desktop review, baseline environmental and social assessment to determine the status of the settlement and filling out an environmental and social risk screening checklist. As a result, the predicted environmental and social impacts of physical planning and land surveying in Ndumanu informal settlement were identified.

The findings of the Environmental and Social Screening indicated that Ndumanu informal settlement is not located in a protected or reserved site. Areas of high ecological value in the settlement were not observed. However, Mount Kenya forest is located 1 Km away from the boundary of the settlement, but would not be affected by the project. The settlement did not have areas which are protected under international, national or local legislation for their ecological, landscape, cultural or other value. In addition, the settlement was not degraded and there were no areas which were subject to pollution or environmental damage, where existing legal environmental standards were exceeded. There were no areas or features of historic or cultural importance which could be affected by the project. Tenure regularization project will not affect groundwater quality, soil or vegetation loss.

The predicted positive impacts include tenure security, employment creation, and improved social amenities, while the negative impacts include structure loss, land-related disputes, exclusion of vulnerable groups, noise and air pollution among others. These projected negative impacts are less severe and can be mitigated through planned actions.

Overall, this screening process determined that the land planning and surveying phases did not require an Environmental Impact Assessment. However, the subsequent sub-component of infrastructure upgrading are expected to have significant environmental implications. This

report therefore recommends that this sub-component should undergo an Environmental Impact Assessment. It is anticipated that these sub-projects may lead to adverse environmental and social effects that are confined to the project area.

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LIST OF ABBREVIATIONS

AFD	Agence Française de Dévelopement
AQI	Air Quality Index
ARAP	Abbreviated Resettlement Action Plan
CAD	Computer Aided Design
CIDP	County Integrated Development Plan
CPCT	County Project Coordination Team
dBA	Decibel
EHS	Environment Health and Safety
EIA	Environmental Impact Assessment
EMCA	Environmental Management and Coordination Act
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESS	Environmental and Social Screening
GIS	Geographic Information System
GoK	Government of Kenya
GPS	Global Positioning System
GRC	Grievances Redress Committee
GRM	Grievances Redress Mechanism
Ha	Hectares
HIV/AIDS	Human Immunodeficiency Virus/Acquired Immune Deficiency Syndrome
IDA	International Development Association
IPF	Investment Project Financing
IUCN	International Union for Conservation of Nature
ISUDP	Integrated Strategic Urban Development Plan

KDSP	Kenya Devolution Support Program
KISIP	Kenya Informal Settlements Improvement Project
KUSP	Kenya Urban Support Program
LPG	Liquefied Petroleum Gas
LPLUDP	Local Physical and Land Use Development Plan
M&E	Monitoring and Evaluation
MDG	Millennium Development Goals
NAMSIP	Nairobi Metropolitan Services Improvement Project
NEMA	National Environment Management Authority
NPCT	National Project Coordination Team
NSP	National Spatial Plan
OP	Operational Policy
PAD	Project Appraisal Document
PAP	Project Affected Person
PLUPA	Physical and Land Use Planning Act
PLWD	Persons Living with Disabilities
PM	Particulate Matter
$PM_{2.5}$	Fine particulate matter of particles up to 2.5
PPE	Personal Protective Equipment
RAP	Resettlement Action Plan
RIM	Registry Index Map
RPF	Resettlement Policy Framework
SDA	Seventh Day Adventist
SDG	Sustainable Development Goal
SEA	Sexual Exploitation and Abuse
SEC	Settlement Executive Committee

SEF	Stakeholder Engagement Framework
SEP	Stakeholder Engagement Plan
Sida	Swedish International Development Cooperation Agency
SMP	Social Management Plan
SPM	Safeguards Procedures Manual
ToR	Term of Reference
UN	United Nations
VMG	Vulnerable and Marginalized Groups
VMGF	Vulnerable and Marginalized Groups Framework
WB	World Bank
WIBA	Work Injuries Benefits Act

CHAPTER ONE: INTRODUCTION

1.1 Background and Context

1.1.1 Background and Perspective on Informal Settlements

Developing countries have implemented housing policies that have not succeeded in providing adequate housing for the informal segments of the urban society. Over time, urban informal settlements have emerged as significant challenges for governments and local authorities, especially in regions such as Asia, Africa, and Latin America. Many residents are unable to secure affordable housing, forcing families to reside in substandard conditions with limited access to sanitation, electricity, and safe drinking water. Furthermore, individuals living in these informal settlements often face insecurity regarding land tenure and rights, living under the persistent threat of eviction.

The situation is further complicated by the rapid growth of many developing towns, which often occurs without adequate planning and essential infrastructure. Residents contend with insufficient infrastructure and uncertain land ownership. Buildings are frequently constructed in areas where the legal framework is unclear, thereby facilitating the growth of informal settlements in developing nations.

UN-Habitat's primary objective is to foster human settlements that are both socially and environmentally sustainable, championing urbanization as a beneficial influence for individuals and communities. This involves promoting inclusive urban development that prioritizes the needs of all residents, particularly marginalized and vulnerable groups. By advocating for equitable access to housing, basic services, and public spaces, UN-Habitat aims to create cities that are not only livable but also resilient to the challenges posed by climate change and rapid urbanization.

1.1.2 Project Background

Kenya Vision 2030, the country's development program, considers the importance of well-functioning cities and metropolitan regions for the country's economic future. The World Bank has contributed to this vision through four programs: The Kenya Urban Support Program (KUSP), the Nairobi Metropolitan Services Improvement Project (NAMSIP), Kenya Devolution Support Program (KDSP), and the Kenya Informal Settlements Improvement Project (KISIP). According to World Bank report on Kenya Informal Settlements Improvement Project (KISIP) 2009, it is estimated that 39 percent of the Kenyan population lived in urban areas where, more than half (54.8 percent) lived in informal settlements. The informal settlements are characterized by; - constrained access to water and sanitation, tenure insecurity, inaccessibility, poor lighting and extensive flooding.

The rapidly growing populations of Kenya's towns, cities and informal settlements have placed additional pressure on basic services, with many lacking accesses to basic water and electricity services. The World Bank, the Swedish International Development Cooperation Agency (Sida) and the Agence Française de Dévelopement (AFD) have signed an agreement with the Kenya Government to improve infrastructure and people's living conditions in informal settlements across Kenyan counties. KISIP complements the National Kenya Slum Upgrading Program, established in 2003, and led by the State Department of Housing and Urban Development. It also supports the efforts by the State Department of Lands and

Physical Planning to introduce planning and strengthen security of tenure in unplanned and poor urban neighborhoods in accordance with the National Land Policy.
The development objective of the Second Kenya Informal Settlements Improvement Project is to improve access to basic services and tenure security of residents in urban informal settlements and strengthen institutional capacity for slum upgrading. The consultancy services to undertake physical planning, cadastral survey and detailed topographical survey of 12 selected informal settlements in Nyeri, Nyeri and Marsabit Counties, Phase IIB – cluster 1 Kenya is aimed at achieving the KISIP 2 development objective. The project will play a key role in contributing to the overall Kenyan Government's goals towards achievement of the Sustainable Development Goals, 2015.

1.1.3 Project Objective

KISIP2 development objective is "**to improve access to basic services and tenure security of residents in participating urban informal settlements and strengthen institutional capacity for slum upgrading in Kenya**"

1.1.4 Project Components

Component 1: Integrated Settlement Upgrading
This component supports settlementupgrading through two main interventions classified under two sub-components:

(i) Subcomponent 1.1: Tenure regularization
Coordinates regularization of tenure for people living on uncontested public lands whose process includes: Development of a local physical and land use development plan (LPLUDP) for the settlement which lays out land parcels and way-leaves for infrastructure like roads, drainage, walkways; Surveying with physical placement of beacons to demarcate the parcels as per theplan; Preparation list of beneficiaries and or issuance of letters of allotment based on the survey plan; and Issuance of titles.

(ii) Sub-component 1.2: Infrastructure Upgrading
Coordinates infrastructure investment portfolio which constitutes of water and sanitation systems, storm water drainage, solid waste collection and settlement sorting, pedestrian walkways, cycle paths, roads, street and security lighting, vending platforms, public parks, and green spaces. It further includes investments related to prevention of crime and violence, including but not limited to community centers.

Component 2: Socio-Economic Inclusion Planning
This component supports community development plans to enhance social and economic inclusion, identifies beneficiaries who fit the eligibility criteria of government programs but areexcluded and connects them appropriately, supports participatory crime and violence mapping, monitors the employment of local labor, carries out community capacity building and awareness raising for various project interventions including community-based solid waste management.

Component 3: Institutional Capacity Development for Slum Upgrading
This component supports institutional and policy development at national and county levels; develops a capacity building plan for national and county levels to implement the

strategy andto develop understanding of slum upgrading processes; also supports technical assistance, training, workshops and learning events, experience sharing and peer-learning activities withother counties, and other capacity building activities.

Component 4: Program Management and Coordination

This component supports activities of the NPCT and the CPCTs related to national and county- level project management and coordination, including planning, surveying, engineering, fiduciary (financial management and procurement), safeguards compliance, monitoring and evaluation (M&E), communication and community development

This consultancy is being implemented under KISIP's first component, sub-component 1.1 which coordinates regularization of tenure for people living on uncontested public lands.

1.2 Deliverables

The environmental and social screening report is one of the listed deliverables under this consultancy. Other deliverables are:

a. An inception report.
b. Base map and environmental status map for each informal settlement.
c. Draft stakeholders' analysis and community engagement report for each informal settlement.
d. Social economic survey report and preliminary list of beneficiaries for each informal settlement.
e. Social and environmental screening report for each informal settlement.
f. Draft Local Physical and Land Use Development Plan for each informal settlement
g. Draft social management plan and/or Resettlement Action Plan (RAP) for each informal settlement
h. Final Local Physical and Land Use Development Plan and a report for each informal settlement
i. Draft survey map and report for each informal settlement.
j. Detailed topographical and engineering map for each informal settlement.
k. Final survey maps and beacon certificates for each informal settlement signed by all beneficiaries.
l. Approved survey plans signed and sealed Registry Index Map (RIM) complete with the area list, approved Local Physical and Land Use Development Plan (LPLUDP) and adopted list of beneficiaries for each informal settlement.
m. GIS data base incorporating both spatial and attribute data on social economic survey, list of beneficiaries, LPLUDP and Survey data for each informal settlement.
n. Final report.

1.3 Project Activities

Environmental and Social Screening report highlights the project activities that will determine the necessity of an Environmental and Social Management Plan (ESMP) or Environmental and Social Impact Assessment (ESIA). Activities of sub-component 1.1 will include data collection, preparation of a base map, preparation of a Local Physical and Land Use Development Plan and issuance of land ownership documents. Activities of sub-component 1.2 will include upgrading of roads, utility lines installations, repairs of and other

construction activities. The environmental impacts of the activities of sub-component 1.2 will be addressed in the specific project EIAs.

CHAPTER TWO: PROJECT SCREENING METHODOLOGY

2.1 Overview

Environmental and social screening is a procedure designed to assess the necessity of conducting an Environmental and Social Impact Assessment for land planning, land surveying, and the enhancement of basic infrastructure in informal settlements. This requirement is outlined in the Second Schedule of the Environmental Management and Coordination Act (EMCA), 2015 Legal Notices No. 31 and 32. The screening process also determines the appropriate level of assessment needed. Consequently, screening serves as a critical decision-making step initiated at the preliminary phases of project execution.

2.2 Significance of the Environmental and Social Screening

The screening process will ensure that the project is implemented in an environmentally and socially sustainable manner and consequently informs whether or not an Environmental Impact Assessment Study should be undertaken.

The screening process will also ensure that the environmental and social risks that could be associated with land planning and survey of Ndumanu informal settlement are remediated.

2.3 Screening Objectives

The primary objective of the screening process was to establish whether tenure regularization of Ndumanu informal settlement falls under the Second Schedule of EMCA, Legal Notice No. 31 and 32. This will determine whether a mandatory impact assessment is required or not (Environmental and Social Management Framework, 2019). The main objectives of the screening were;

i. To examine the potential positive and negative environmental and social impacts of land planning and surveying in Ndumanu informal settlement, Nyeri County.
ii. To evaluate the land planning and survey, environmental and social impacts' magnitude and scope.
iii. To determine the feasibility of the tenure regularization process.

2.4 Screening Methodology

Screening methodology refers to the systematic approach used to predict the environmental and social impacts of the project's interventions on an informal settlement. The approaches used in this environmental and social risks screening process are discussed below.

2.4.1 Desktop Review

A literature review was undertaken to understand the institutional, policy, and legislative framework relevant to the project, including an indication of the key applicable provisions, policy guidelines, and standards that will have to be addressed in the environmental and social screening report. The report was prepared in line with the Terms of Reference (ToR), KISIP2 Environment and Social Management Framework (ESMF), KISIP2 Resettlement Policy Framework (RPF), Project Operations Manual, World Bank's Project Appraisal Document,

World Bank's Operational Policies (OP4.01, OP4.11, OP4.12), World Bank Environment Health and Safety (EHS) Guidelines, Policy, legal, regulatory, and institutional frameworks for environmental and social management in Kenya and Nyeri County Integrated Development Plan (CIDP 2023-2027) among other provisions.

2.4.2 Baseline Environmental and Social Survey

A site visit was undertaken in Ndumanu Informal Settlement on 26th and 27th November 2024 during which a site inspection was carried out with a focus on a wide range of environmental and social attributes including physiography, soil, water resources, vegetation land use, and the infrastructure of the settlement. The site inspections were aimed at determining the preliminary baseline of environmental and social status of the settlement and views regarding the tenure regularization activities. The assessment entailed activities which were;

i. Assessment of Ndumanu informal settlement's topography which was undertaken during site inspection using visual observation.
ii. Evaluation of the Ndumanu informal settlement's drainage using existing literature followed by ground truthing.
iii. Vegetation survey conducted at Ndumanu informal settlement based on visual observation.
iv. Visual observations of Ndumanu informal settlement characteristics to ascertain the housing conditions, available physical and social infrastructure, non-motorized transportation, economic activities, land uses, community identity, and the environmental situation.

2.4.3 Stakeholder Engagement and Consultation

The need for public involvement in project development is enshrined in the Constitution of Kenya, 2010. This requirement is also provided for in the EMCA, 1999 (Amended 2015), and is one of the guiding principles of the National Environment Policy, 2013.

Stakeholder consultations were undertaken on 20th November 2024 with an objective of informing the community and other stakeholders about what the project entails and its benefits. Consultation was undertaken to collect views, comments and concerns and finally seeking consensus and community consent on the project.

2.4.4 Environmental and Social Screening Checklist

The Environmental and Social Management Framework (ESMF) requires that all potential projects funded under KISIP2 be screened for environmental and social impacts using the screening checklist. The screening checklist is a tool used to evaluate the potential environmental and social risks associated with a project.

The Resettlement Policy Framework (RPF) provides a comprehensive framework outlining policies, procedures, entitlements, eligibility criteria, grievance redress mechanisms, stakeholder and Project Affected Persons (PAPs) engagement, and institutional arrangements to manage and address displacement impacts resulting from project implementation. The guidelines outlined in the Resettlement Policy Framework (RPF) will be implemented during infrastructure improvements. These upgrades may result in the displacement of structures and livelihoods.

An environmental and social screening checklist serves as a mechanism to assess the possible environmental and social repercussions of projects, as well as to ascertain if they will activate applicable safeguard policies established by the World Bank. This tool is utilized to screen, categorize, and evaluate project activities throughout the preparation phase.

An environmental and social risks screening checklist was the main tool used for identifying environmental and social impacts in Ndumanu informal settlement which was filled during the site visit. The checklist examined issues ranging from World Bank safeguard policies, Kenyan policies and legal frameworks, and settlement aspects such as environmentally sensitive areas, project construction and operation phase effects, impacts on water resources, socio-economic implications, and occupational health and safety within the Ndumanu boundary.

The data gathered from the screening methodology was analyzed to identify the potential environmental and social impacts which were then assessed for their significance to the environmental and the social well being of the residents of Ndumanu informal settlement.

CHAPTER THREE: LEGAL, INSTITUTIONAL AND REGULATORY FRAMEWORK

3.1.Overview

This chapter outlines various legal policies, KISIP 2, and World Bank instruments that directly impact the delivery of the assignment. In particular, it highlights their objectives and relevant applications, which will be important for the execution of the assignment. Table 1, 2 and 3 describes the legal and institutional regulatory frameworks and their applicability in this project.

Law/Regulation	Objective	Application to 1.1
The Environmental Management and Coordination Act (EMCA), 1999 (Amendment 2015)	The Act establishes a framework for realization of the constitutional right to a clean environment and defines the state's role in ensuring environmental sustainability.	• This consultancy necessitates the preparation of environmental and social screening reports to identify potential environmental and social impacts mandating mitigating measures as well as promoting sustainable environmental management in each of the informal settlement. • The settlement's LPLUDP shall ensure a clean and healthy environment by provision of an efficient solid and liquid waste management plan. • Public participation will be incorporated throughout the project cycle.
EMCA 1999 (EIA/EA Regulations 2019)	The legal notice No. 31 and 32, provides guidelines on classification of the projects into low risks, mediums risk and high risk. The legal notice No. 32 also provides guideline on contents of Summary Project Report, Comprehensive Project report and study report	• Based on the identified risks, physical planning and surveying activities will be managed through implementation of mitigation measures. Therefore, Environmental and Social Screening Assessment is adequate.
The Sustainable Waste Management Act, 2022	The Act creates a legal and institutional framework to sustainably manage waste, ensuring the fulfillment of constitutional provisions regarding the right to a clean and healthy environment.	• The LPLUDP to be prepared as part of the consultancy assignment shall provide a detailed analysis of solid waste management in each selected informal settlement. • The factors under analysis shall include but not limited to; ➢ Existence of solid waste disposal facilities in the settlements. ➢ Establish sustainable solid waste management practices in the settlements. ➢ Provision of on source solid waste sorting facilities and along transportation corridors in the settlements. • Establish ways and means of how solid waste in the settlement could be an opportunity for employment.

Law/Regulation	Objective	Application to 1.1
The Land Act, 2012	The Land Act of 2012 operationalizes Chapter Five of the Kenyan constitution on land.	• The consultancy entails tenure regularization of the 12 selected informal settlements that will be addressed through preparation of LPLUDPs, cadastral surveying and land registration. • The project will ensure that the informal settlements are well planned, surveyed, and serviced before allocating and issuing lease certificates to the beneficiary residents.
The Land Registration Act, 2012	The Land Registration Act of 2012, aligns with Article 61 of the Kenyan constitution, which classifies land as public, private, or community- owned. This legislation facilitates the registration of land under above aforementioned classifications.	• The project will emphasize to achieve the goal of security of land tenure in the settlements, through preparation of LPLUDP and survey. • The consultant, working in collaboration with the Settlement Executive Committee (SEC) and residents, will develop a comprehensive List of Beneficiaries (LOB). This list may include joint beneficiaries for a single parcel. The consultant shall guide joint owners on the process of agreeing on the recipient of the eventual certificate of lease, emphasizing the consequences of non-agreement. • The informal settlement LPLUDP shall classify public, community and private land for purposes of issuance of certificate of title or lease as the case may be
The National Lands Commission Act, 2012	The Act gives effect to Article 67 of the Constitution, establishing the National Land Commission (NLC) and outline of its function.	• The project involves land tenure regularization and issuance of land ownership documents on government owned land therefore national land commission as the custodian will be instrumental in reservation of the settlements for purposes of planning and surveying. • As one of the project outputs, Local Physical Land Use Development Plans will be prepared in accordance with the provisions of National Land Commission Act which guides land use planning in Kenya. • The consultant in collaboration with the counties' NLC offices will encourage local land dispute resolution mechanisms to minimize lengthy court injunctions that may impede project progress by utilizing the elected Grievance Redress Committees (GRCs) to address land-related conflicts thereby promoting the project's smooth implementation within the framework of land management overseen by the NLC

Law/Regulation	Objective	Application to 1.1
The Physical and Land Use Planning Act (PLUPA), 2019	To guide on how the Local Physical and Land Use Development Plans will be prepared.	• This consultancy entails tenure regularization on uncontested public land. The requirement of the Land Act 2012 section 13 demands that for public land to be allocated it has to be planned, surveyed and serviced. The planning envisaged by the land Act is as provided for by the Physical and Land Use Planning Act No. 13 of 2019. • The preparation of the plan from the inception to approval will be guided by the provision of this Act together with other relevant legislative framework like the Survey Act, EMCA, the Urban areas and cities Act among others. • It is worth noting that because of the peculiarities of the participating informal settlements, the normal conventional planning may not be applicable and a lot of trade-offs may be employed by way of taking into account adaptive planning techniques in order to reduce displacement or disruption. Care will be taken to ensure that security and public health standards are not compromised in the process.

Table 1: National Laws and Regulations

Policy	Objective	Application to 1.1
Sustainable Development Goals (SDGs)	The Sustainable Development Goals (SDGs), adopted in 2015, chart a global road-map for a better future. Replacing the Millennium Development Goals, these 17 ambitious goals, with 169 targets, address critical challenges like poverty, hunger, and climate change.	• The SDGs are relevant in this assignment in that they provide global perspective for addressing various social, economic and environmental challenges faced by residents of informal settlements. • This assignment involves tenure regularization through physical planning and survey of selected informal settlements. The main aim is to regularize land tenure, improve access to basic infrastructure and services with aim of improving the quality of life of the residents so the goals will come in handy in articulating these needs. • The SDGs will be mainstreamed in the plan proposals in line with the principle of subsidiarity. • SDGs advocates for inclusivity. Informal settlements are epitomes of inequality in this respect in the tenure regularization process the disadvantaged, minority and women will not be discriminated.
Kenya Vision 2030	It serves as the country's long-term development blue print, aiming to transform Kenya into a newly industrialized middle-income country by the year 2030.	• The preparation and issuance of land ownership documents ensures property rights which is crucial for addressing challenges in informal settlements. • Local Physical and Land Use Development Plans will incorporate strategies to enhance sustainable urban development and affordable housing options as stressed in the vision. • Tenure regularization will incorporate strategies to enhance a clean, safe and sustainable environment to access to water and sanitation services.
Sessional Paper No. 2 of 2016 On the National Slum Improvement and Prevention Policy (NSUPP)	The overall objective of this policy is to promote, secure and protect dignified livelihoods of the poor living and working in the slums by strategically integrating them into the social, political and economic framework in line with the constitution.	• The informal settlements local physical and land use development plans to be prepared under this assignment are aimed at providing land tenure security and improvement of the settlements through provision of social and physical infrastructure.

Policy	Objective	Application to 1.1
National Policy for Disaster Management in Kenya, 2009.	The Policy seeks to enhance and maintain the resilience of vulnerable communities to hazards by diversifying their livelihoods and bolstering coping mechanisms.	• The policy emphasizes the importance of putting up in place measures that reduce risks associated with disasters such as land-use planning initiatives and public awareness campaigns. Tenure regularization process will incorporate disaster management strategies.

Table 2: Policy Framework

Instrument	Objective	Application to 1.1
OP 4.01 Environmental Assessment. Category A - likely to have significant adverse environmental impacts that are sensitive, diverse, or unprecedented. Category B – same environmental impacts as Category A but are less adverse than those of Category A projects. Category C - likely to have minimal or no adverse environmental impacts. Beyond screening, no further EA action is required. Category FI - involves investment of Bank funds through a financial intermediary in subprojects that may result in adverse environmental impacts.	To ensure environmental soundness and sustainability.	Categories A, B, C Are applicable to component 1.1. The Operational Policies advocate for protection of the environment against adverse human activities, they promote sustainable use and management of resources and utilization of land. Tenure regularization will have minimal impact to the natural environment. It will ensure clean and sound environment.
OP 4.12 Involuntary Resettlement, 2013.	To ensure minimal disturbances. To ensure minimal displacement and related social conflicts. To ensure harmonious relocations.	The Operational Policy promotes participatory planning approach which encourages minimal and involuntary resettlement. The policy shall be adopted in sorting harmony in cases of structures encroachment into wayleaves, roads or adjacent public spaces. It will also ensure maximum utilization of the settlement committee in solving related resettlement matters. The instrument shall be mainly applicable in bringing harmony aiming at achieving its objective in the informal settlement.

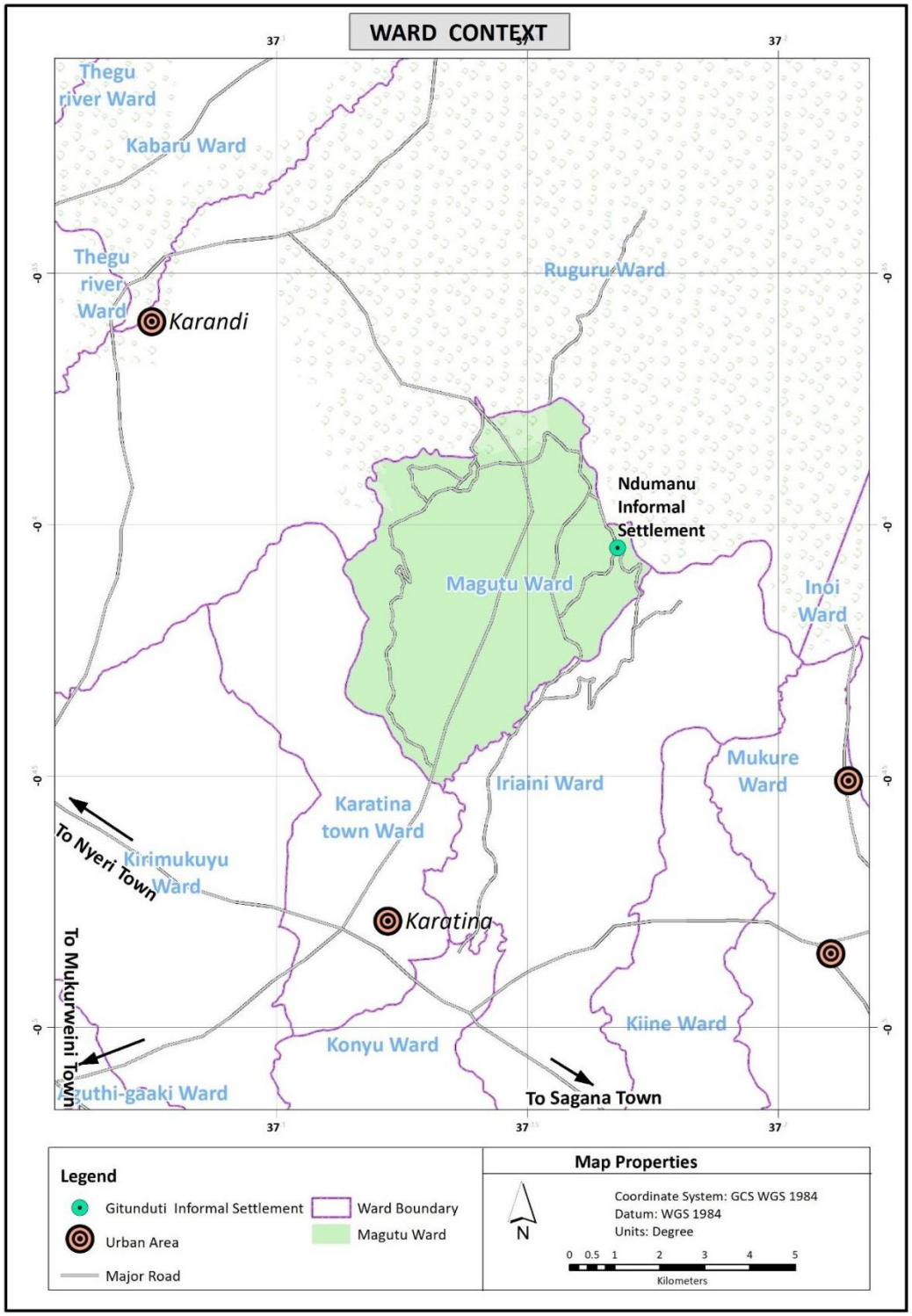
Instrument	Objective	Application to 1.1
OP 4.10 Indigenous People;	To ensure social inclusivity and protection. To ensure adequate consultation and participation. To ensure equal rights and voices.	The Operation Policy is relevant in the promotion of indigenous people equal rights to access land and sustainable utilization of resources. The informal settlement is inhabited with the indigenous community, both genders and with varying age limits.

Table 3: KISIP 2 and World Bank Instruments

CHAPTER FOUR: SETTLEMENT BASELINE INFORMATION

4.1. Settlement Location and Size

Ndumanu informal settlement is located off the Itundu-Ragati Road in Magutu ward, Mathira East sub-county of Nyeri County, Kenya. The ward borders Ruguru Ward, Karatina Town Ward, Iriani Ward and Kirinyaga County. It measures approximately 8.5 Hectares.



Map 1: Location of Ndumanu Informal Settlement in the Ward Context

4.2. Bio-Physical Baseline

Biophysical information encompasses the biological and physical characteristics of the environment. The physical characteristics include landforms, types of vegetation, climate conditions, and soil types. The biological component illustrates the connections between the physical environment and living organisms.

4.2.1. Geology and Soils

The soils of Ndumanu are generally deep well-drained basic red-volcanic soils, referred to as *Andisols*. The soils are characteristically fertile for arable agriculture. They have closely packed particles and lack sand. They are sticky during wet seasons making weather roads unsustainable year-round.



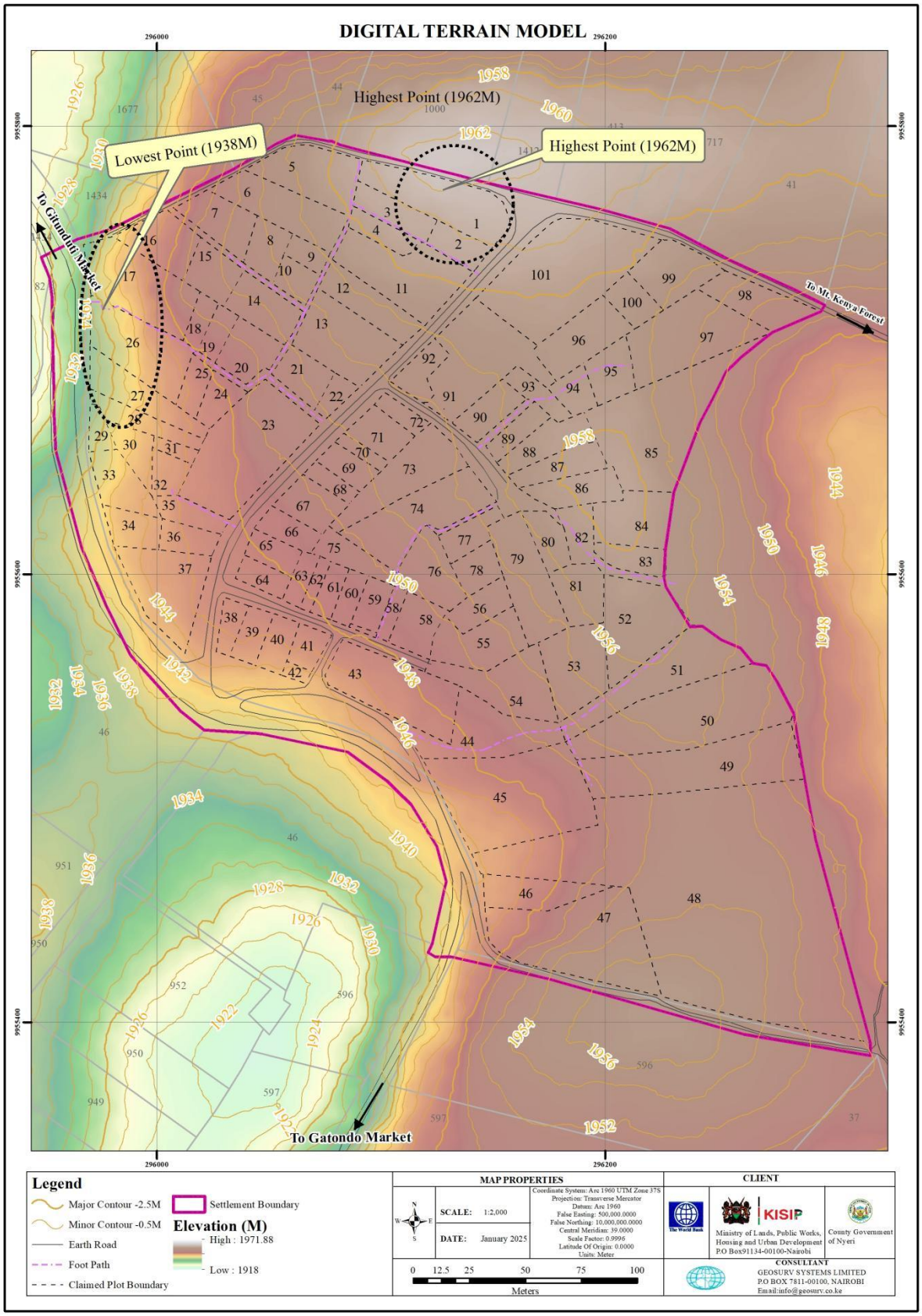
plate 1:Soils in Ndumanu
Source: Field Survey, 2024

4.2.2. Hydrology

The type of precipitation experienced in the settlement is rainfall. The rainfall is experienced in two seasons. The rainfall is evenly distributed across the settlement. There are no rivers, streams, lakes or wetlands in the settlement.

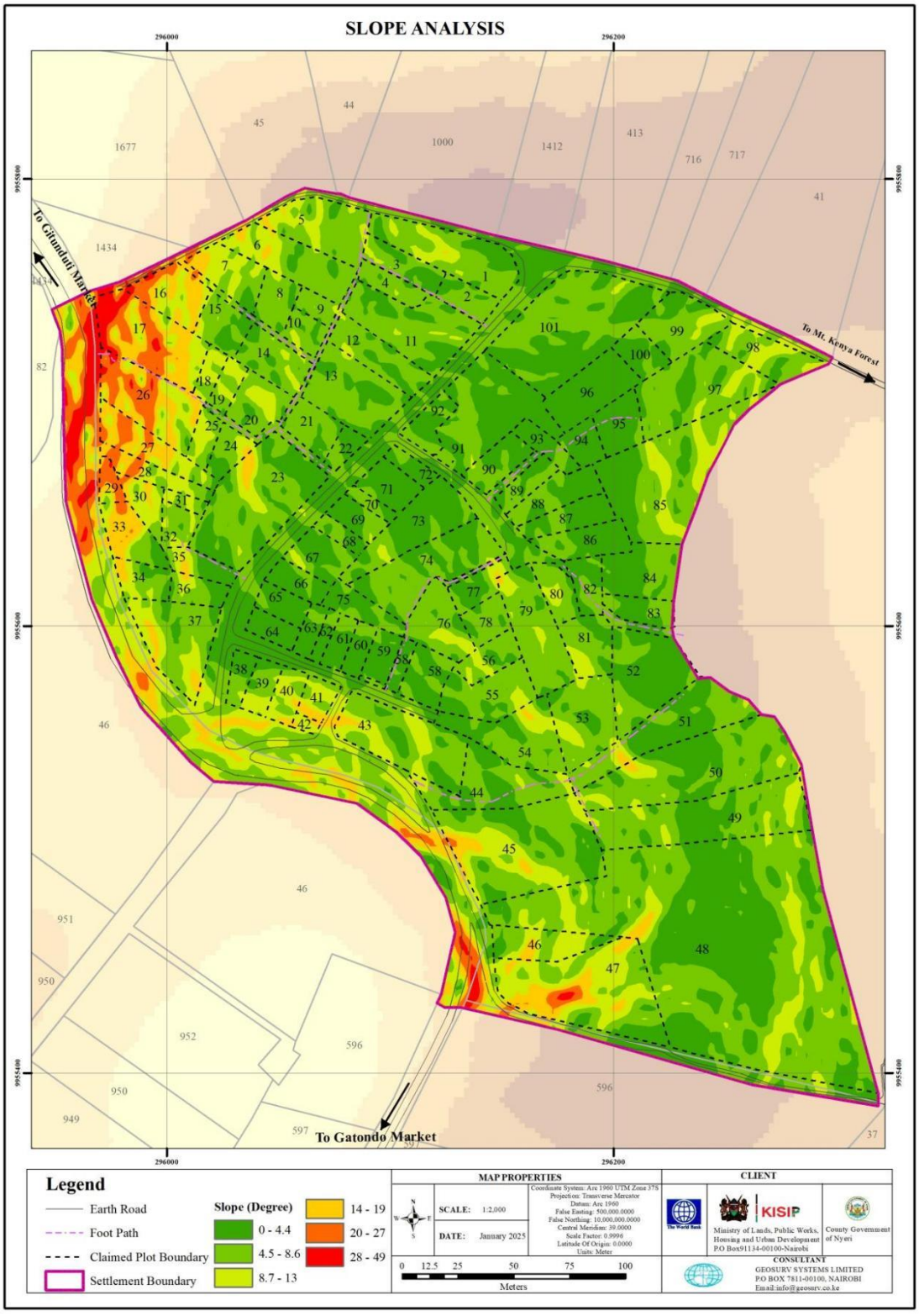
4.2.3. Topography

Topographic analysis involves the assessment and interpretation of the physical attributes and characteristics of a landscape or terrain. The settlement had a gently sloping gradient with some parts being steeply. The steep slopes were used for agriculture while residential premises were on the gently and flat terrain.



Map 2: Digital Elevation Model for Ndumanu

The settlement sloped from the East towards the West direction. The highest point was at an altitude of 1962 meters above sea level while the lowest point was at an altitude of 1938 meters above sea level.



Map 3: Slope Analysis

4.2.4. Climate

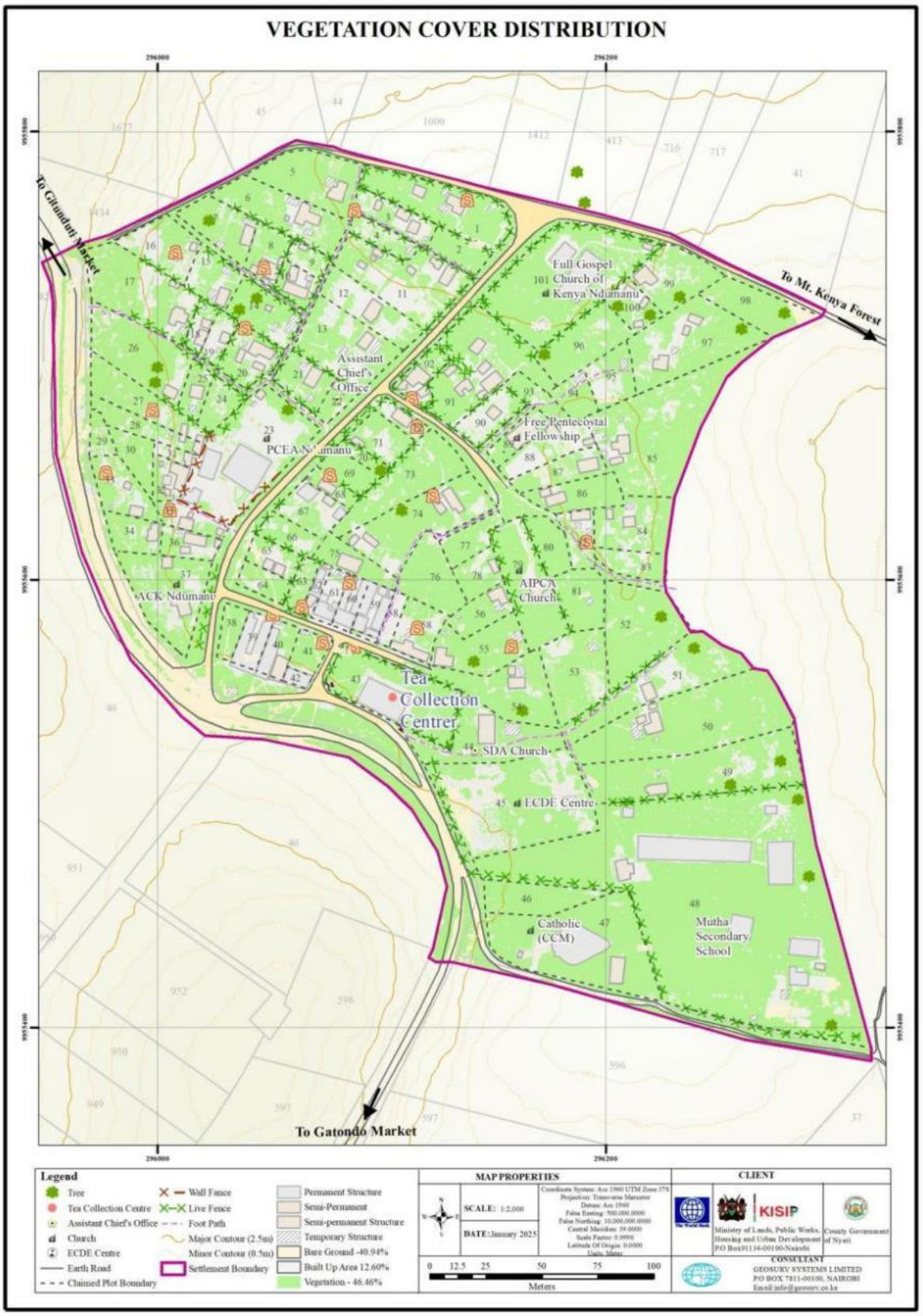
Climate is defined as the weather conditions of an area over a long period of time. The climate of Ndumanu informal settlement is influenced by the climate of Magutu ward which is a cool and wet climate. The settlement experiences two rainy seasons, the long rains occur between the months of March to May and the short rains occur between the months of October to November. The temperature ranges from 19 degrees Celsius to 29 degrees Celsius.

Climate change can cause the settlement to be susceptible to extreme weather conditions such as high temperatures, flooding among others. Addressing climate change impacts in the settlement calls for integrated strategies that boost resilience, improve basic infrastructure, increase access to essential services, and empower the residents to adapt to changing climatic conditions.

4.2.5. Vegetation

Ndumanu informal settlement had natural and planted vegetation. The predominant vegetation was tea plants and grass. Other vegetation included blue gum trees, crops like nappier grass, maize and vegetables among others. According to the International Union for Conservation of Nature (IUCN) Red List for Kenya, all of the species spotted in the settlement are rated as 'least concern'.

The settlement did not have ecologically sensitive areas that required protection or conservation.



Map 4: Vegetation Cover Distribution



plate 2: Vegetation Types
Source: Field Survey, 2024

4.3. Settlement Characteristics

4.3.1. Population

The population refers to the total number of people living in a specific area at a given time. Population dynamics is crucial for various aspects of societal planning and development such as provision of social amenities, essential services and labour force. Ndumanu informal settlement has a population of approximately 600 people according to the KISIP ToR, 2024.

4.3.2. Land Tenure

Ndumanu informal settlement is located on uncontested public Land Registration Number Magutu/Gatei/514. It is registered under the County Government of Nyeri.

4.3.3. Land Uses

Land use refers to the human activities and functions carried out on a particular piece of land. It encompasses the various ways in which land is utilized, managed, and modified for different purposes. The settlement was characterized by residential, agricultural, public purpose and commercial land uses.

4.3.4. Housing Typology

Housing typology refers to the classification or categorization of different types of housing based on various characteristics such as architectural style, building form, size, layout, and intended use. It provides a framework for understanding the diversity of housing options available within a given context.

Ndumanu informal settlement had different housing typologies constructed with different materials and styles. The housing typologies included row housing and bungalows. The houses were constructed mainly by use of wood. Other construction materials were stone iron sheets. The housing condition was so poor in the settlement where most of the houses were in a dilapidated state.



plate 3: Housing Typology

Source: Field Survey, 2024

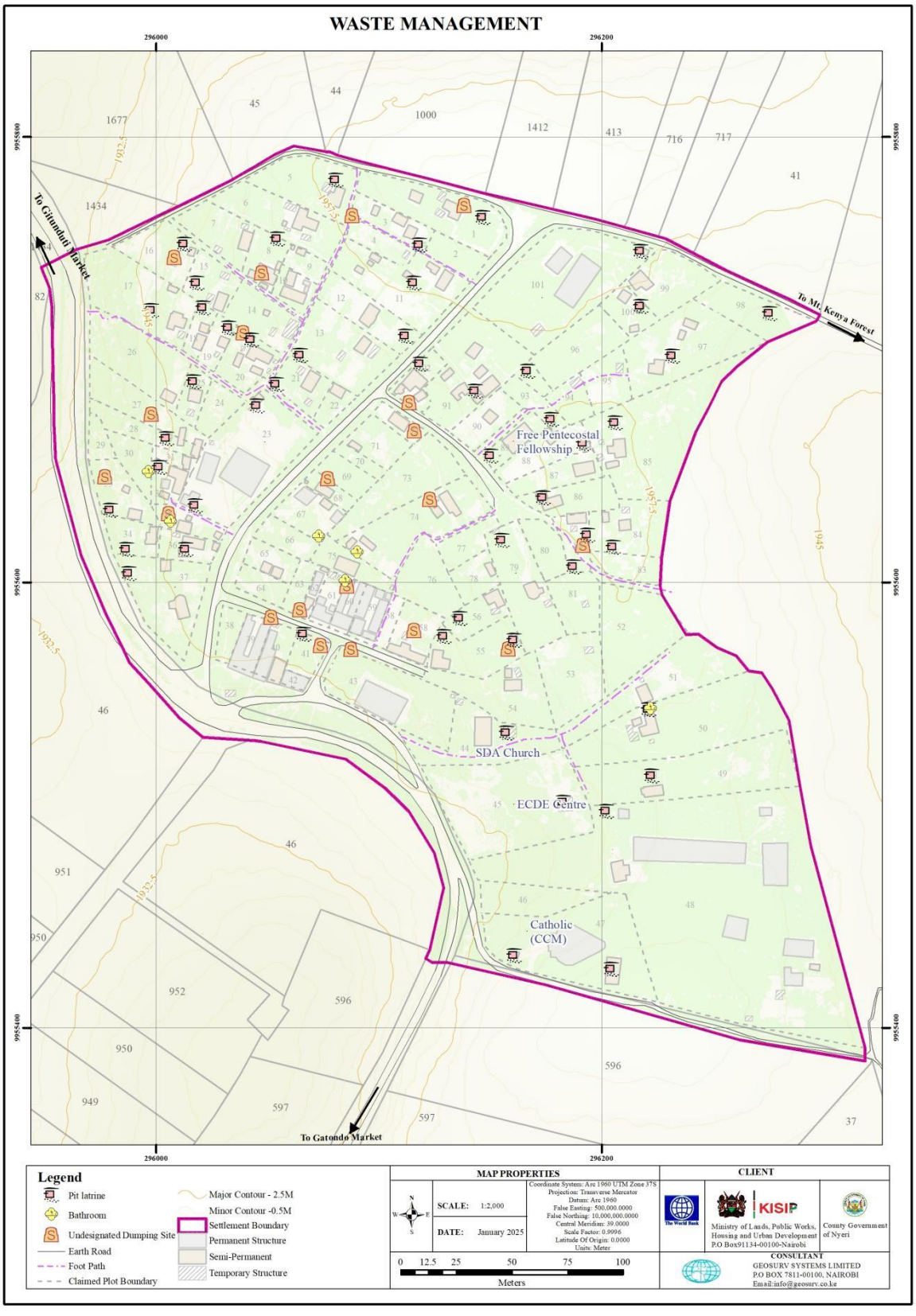
4.3.5. Solid Waste Management and Sanitation

Solid waste management is the process of handling and disposing of waste produced by human activities in a safe, environmentally friendly, and sustainable manner. It involves the collection, transporting, sorting, treating, and disposing of solid waste in a way that minimizes the negative effects on public health and the environment.

Various types of solid wastes were generated in the settlement where the common ones included household wastes such as plastics and papers. Other solid wastes included glass bottles, children dippers, tea leaves remains and clinical wastes among others.

Ndumanu Settlement lacked designated solid waste services. Most of the homesteads in the settlement dumped their waste indiscriminately along the roadside and in their gardens

Improper management of solid waste in Ndumanu poses a significant risk to public health. The presence of uncontrolled waste disposal sites draws flies, exacerbating environmental deterioration.



Map 5: Solid and Liquid Waste Disposal Sites



plate 4: Solid Waste Disposal
Source: Field Survey, 2024

4.3.6. Liquid Waste Management

Sanitation facilities encompasses toilets, septic systems, sewage systems, and wastewater treatment facilities, which handle wastewater generated from toilets, bathrooms, and kitchens. Wastewater comprises human excreta, kitchen refuse, and water from bathing facilities.

Ndumanu settlement had no conventional wastewater disposal and treatment system. Domestic waste water flowed freely in the homesteads. The waste water was generated from activities such as laundry, hand washing, bathing, and dish washing. Laundry and dish washing were done outside homes where water was discarded to the open ground. Most households in Ndumanu Settlement lacked access to modern toilet facilities, which are crucial to preventing open defecation instead they use pit latrines to dispose of their human waste. Bathrooms were located outside the houses. Some residents had dug trenches in their homes to dispose their household waste water.Incorrect handling and management of wastewater and human waste in the settlement increased the danger of developing water-borne diseases.



plate 5: Pit Latrines and Bathroom Condition
Source: Field Survey, 2024

4.4. Physical Infrastructure

Physical infrastructure pertains to the concrete assets, facilities, and systems that underpin economic, social, and environmental activities within a community. It includes a diverse array of constructed structures, networks, and facilities that deliver vital services and facilitate numerous functions and interactions.

4.4.1. Water Sources and Supply

Water supply infrastructure encompasses the systems and facilities required to provide clean and safe water to communities for various purposes. The main water supplier for Ndumanu informal settlement was Kihore Water Project**.** There were two main taps in the settlement where the residents collected water. Some residents had water tanks for storage of water.



plate 6: Water Infrastructure
Source: Field Survey, 2024

4.4.2. Roads

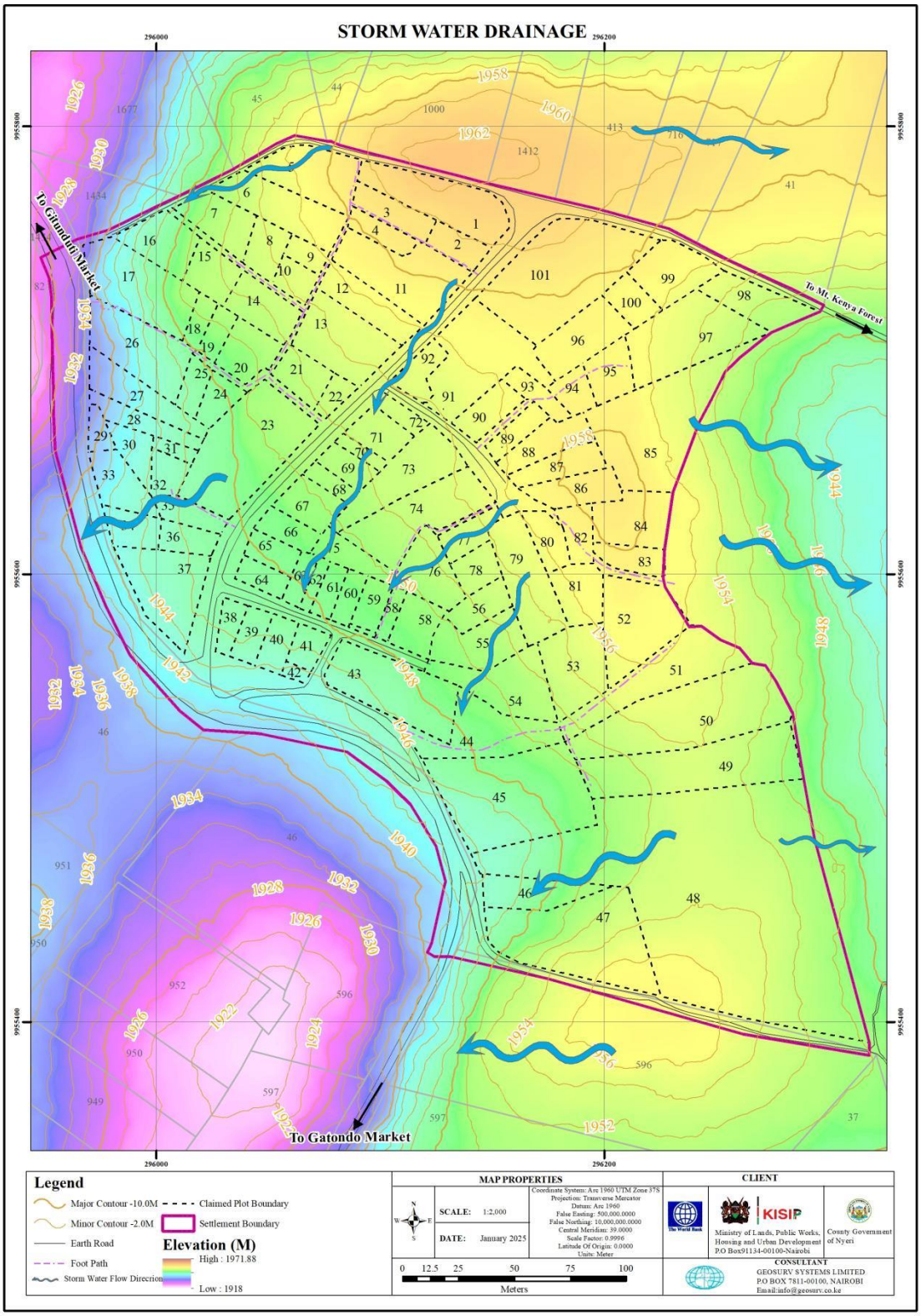
The settlement is mainly accessed from the Itundu-Ragati Road. Other access roads in the settlement were of earth and murram in nature and lacked proper drainage systems leading to soil erosion.



plate 7: Nature of Roads
Source: Field Survey, 2024

4.4.3. Surface Drainage

Surface drainage refers to the removal of excess water from the land surface. Surface drains which play a key role in managing runoff and flood management were lacking in Ndumanu. The access roads had no drainage channels.



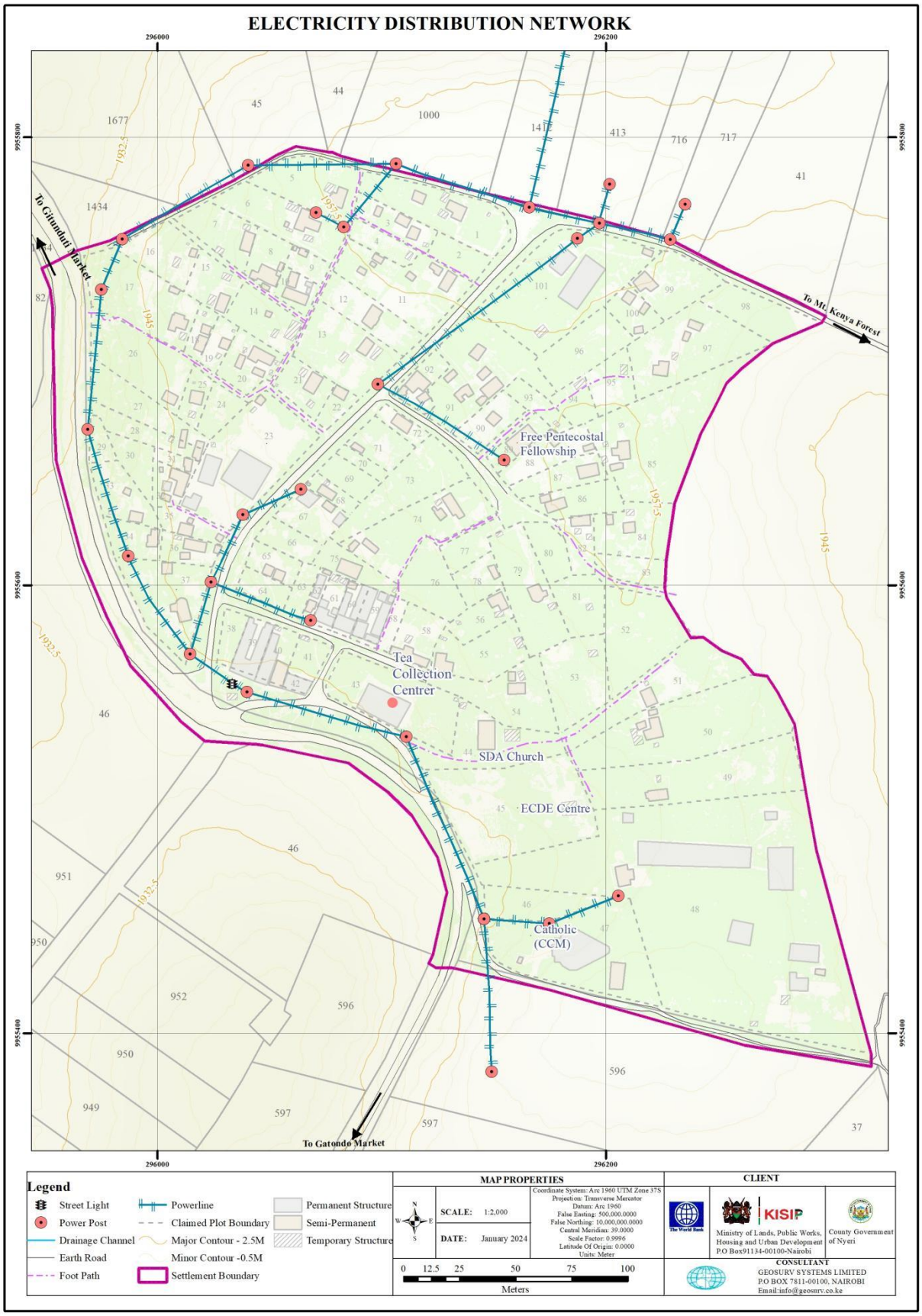
Map 6: Storm Water Flow Analysis



plate 8: Poor Drainage System
Source: Field Survey, 2024

4.4.4. Energy Sources

An energy source is the origin from which energy can be obtained to provide heat, light, or power. The main source of energy for cooking and lighting used by Ndumanu residents was firewood. Other sources included kerosene, charcoal and electricity. The energy sources in the settlement were not sustainable and environmental friendly.



Map 7: Electricity Distribution in Ndumanu



plate 9: Energy Source
Source: Field Survey, 2024

4.5. Social Infrastructure

Social infrastructure refers to the physical structures and facilities that support social activities, interactions and services within a community. The settlement had a number of social infrastructures such as Joymart ECDE Center, Ndumanu Dispensary, Ndumanu Police Post, ACK Church, Ndumanu Catholic Church, Chief and Assistant Chief's Offices, Ndumanu Social Hall, and Ndumanu Tea Buying Center among others.



St. Joseph's Catholic Church



Full Gospel Church



Muthea High School



PCEA Ndumanu Church



Chief's Office

plate 10: Social Facilities
Source: Field Survey, 2024

CHAPTER FIVE: ENVIRONMENTAL AND SOCIAL SCREENING FINDINGS, IMPACTS PREDICTION ASSESMENT AND MITIGATION MEASURES

5.1. Environmental and Social Screening Findings

The screening exercise is a decision-making process that is initiated during the early stages of project implementation. Environmental and social risks are described as the potential changes to the existing condition of the environment caused by human activities. Risks may be either direct or indirect, long-term or short-term, and wide-spread or local in terms of effect scope. The consultant was mandated to identify both the positive and negative impacts of land planning and land surveying in Ndumanu informal settlement. The screening exercise was to determine whether the project interventions would require full environmental and social impact assessment to be done or not.

5.2. Environmental Screening Findings

1. Ndumanu informal settlement is not located on uncontested public land.
2. Ndumanu informal settlement is not located in a protected or reserved site. Mount Kenya forest is located one Kilometer away from the boundary of the settlement but it will not be affected by the project.
3. There are no areas of high ecological values in the settlement.
4. Mount Kenya forest one kilometer from the settlement is an area protected under national legislation for their ecological value. This area however will not be affected by the project.
5. The settlement did not have areas or features of historic or cultural importance which could be affected by the project.
6. The project will not result in loss of green field land in the settlement since the settlement was previously developed for residential, administrative, religious and agricultural purposes.
7. There were steep slopes around the settlement which could limit some infrastructural developments.
8. Tenure regularization will not affect groundwater quality, soil or air quality in or around the settlement.

5.3. Social Screening Findings

1. Tenure regularization will promote social stability by minimizing land-related conflicts, disputes and forced evictions. Establishing clear land tenure systems will create a foundation for resolving disputes and encouraging peaceful coexistence within settlements.
2. Tenure regularization will empower marginalized groups by granting them legal recognition and safeguarding them against forced evictions and discrimination.
3. The project will improve existing infrastructure and propose new upgraded infrastructure.
4. Tenure security will trigger more development in the settlement and this will bring changes in existing land uses. These changes will have significant impacts on population dynamics and the demand for basic services.

5.4. Environment and Social Risks Identification and Ranking

Environment and social risks identification and analysis was done using the Leopold matrix method which was developed by Luna Leopold, an influential environmental scientist. This method is used for systematically identifying and analyzing environmental impacts associated with proposed projects or activities. The matrix essentially creates a grid format, where project activities are listed along one axis, and environmental factors or risks are listed along the other axis. This grid allows for a structured analysis of how each project activity may interact with various environmental aspects, helping to identify potential risks, impacts, and mitigation measures.

5.4.1. Risk Rating Variables

Environmental and social risks are described as the potential changes to the existing condition of the environment caused by human activities. Risks may be either direct or indirect, long-term or short-term, and wide-spread or local in terms of effect scope.

Timing	***Short-term:*** impacts limited to the duration of the activity. (zero up to 1 year)	***Long-term:*** impacts that may last for longer periods. (more than 1 year)	***Permanent:*** impacts that cause a permanent change in the baseline conditions.
Extent	***Local:*** impacts within local communities.	***Regional:*** impacts affecting a wider area or socio-economic asset of importance going beyond the communities	***National:*** impacts extending to the national level, or affecting assets of national importance.
Probability	***Likely:*** past experiences in similar projects give some substantiated evidence that the impact could be frequent.	***Possible:*** could occur but is not frequent	***Unlikely:*** very rare in the country.
Intensity	***Low to negligible:*** limited impacts that do not lead to any long-lasting change.	***Medium:*** impacts causing change but not affecting the core livelihood of affected people or the main structure of the socio-economic activities.	***High:*** impacts causing a complete change of livelihood; or change/destruction of the assets.

Table 4: Risk Rating Variables

5.4.2. Definitions of the Impact Significance Assessment Rating

The significance of the project's impacts may be negligible, low, medium or high as defined on the table below.

Negligible	Negligible magnitude impacts on low or medium sensitive environmental, and socio-economic resources. The impact is hardly distinguishable from background conditions and expected development in a no-project situation.
Low	Low impact on medium or low sensitive environmental, biological, and socio-economic resources, medium impact on low sensitive resources. The negative impact can still be minimized to a negligible level through best practices and mitigation measures
Medium	Low magnitude impact on highly sensitive environmental, biological, and socio-economic resources, medium magnitude impact on medium sensitive resources, and high magnitude impact on low sensitive resources. For negative impacts, mitigation measures are required and, as needed, compensation measures.
High	High or medium magnitude impact on highly sensitive environmental, biological and socio-economic resources. Negative impact must be avoided, mitigated, or compensated to an acceptable level through implementation of specific mitigation or compensation measures.

Table 5: Definition of Environmental and Social Risk Ratings

5.4.3. Rating Criteria for Environment and Social Risks

The timing, extent, probability, intensity, Weighing Factor, Severity Rating and mitigation efficiency of the project risks are analyzed as shown on the table below.

Timing		**Extent**		**Probability**		**Intensity**		**Weighing Factor (WF)**		**Severity Rating (SR)**		**Mitigation Efficiency**	
Short term	1	Local	1	Likely	1	Low to negligible	1	Low to negligible	1	Low to negligible	> 32	Low to negligible	1
Long term	2	Regional	2	Possible	2	Medium	2	Medium	2	Medium	33 – 65	Medium	2
Permanent	3	National	3	Unlikely	3	High	3	High	3	High	66 - 100	High	3

Table 6: Environmental and Social Risk Rating Criteria

5.4.4. Environment Risk Significance Rating

The significance of the predicted environmental and social risks are rated as shown in the table below.

Risk	Rating Factors				Weighting x Weighting Factor	Significance Rating
	Timing	Extent	Probability	Intensity	(T+E+P+I) x WF	
Land related disputes	Short term - 1	Local - 1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Political interference	Short term - 1	Regional - 2	Likely - 1	Medium - 2	(1+2+1+2) x2= 12	Low
Conflicts among neighbours	Short term - 1	Local -1	Possible - 2	High - 3	(1+1+2+3) x2= 14	Low
Exclusion of vulnerable groups	Short term - 1	Local -1	Possible -2	Medium -2	(1+1+2+2) x2= 12	Low
Inequitable land distribution	Short term - 1	Local - 1	Likely – 1	Low to negligible - 1	(1+1+1+1)2= 8	Low
Loss of community identity	Short term - 1	Local - 1	Likely – 1	Low to negligible - 1	(1+1+1+1)2= 8	Low
Loss of structures: Fully or partially.	Short term - 1	Regional - 1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Increased dust & exhaust emissions	Short term - 1	Local - 1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Increased noise and vibration	Short term - 1	Local - 1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Increased water pollution	Short term - 1	Local - 1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low

Increased generation of solid waste	Short term - 1	Local - 1	Likely - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Increased soil disturbance and erosion	Short term - 1	Local - 1	Possible – 2	Medium - 2	(1+1+2+2) x2= 12	Low
Reduced public health and safety	Short term – 1	Local -1	Possible - 2	Medium - 2	(1+1+2+2) x2= 12	Low
Workers safety	Short term - 1	Local - 1	Possible - 2	Medium – 2	1+1+2+2) x2= 12	Low

Table 7: Environmental and Social Risk Significance Rating

5.4.5. Predicted Positive Impacts

a) **Tenure security:** This will provide the residents of Ndumanu with land ownership documents hence legal recognition as the land owners. The ownership documents safeguard the residents against evictions while promoting developments such as housing and infrastructure. The documents can also facilitate access to financial services enabling the residents to engage in income-generating activities thus improving their standards of living.

b) **Improved access to basic services:** Once Ndumanu's residents are legally recognized, they will be eligible for essential services such as water, sewer lines, electricity, education facilities, health care, and recreation areas which are inadequate at present. The essential services will improve the residents living conditions, promoting good health and sustainable development.

During the upgrading of basic infrastructure potential benefits include;

- **Creation of employment:** The construction phase of the project will have economic benefits for the local population. Employment should prioritize the residents of Ndumanu settlement more so where personnel at lower skill levels are required. It is envisaged that employment will have a direct injection of money into the local economy, with persons working on the project spending some of their wages in the local area. This will increase business activities for local entrepreneurs supplying goods to such persons.
- **Improved road access:** accessible roads are crucial for ensuring efficient transportation of goods and services. They enable residents of Ndumanu to connect with essential facilities like schools, healthcare, and employment opportunities. Additionally, the roads contribute to emergency response capabilities, ensuring timely access to medical services and disaster relief.
- **Pedestrian walkways:** They will contribute significantly to the safety and mobility of Ndumanu residents by enhancing interconnectivity within the settlement, allowing residents to efficiently navigate and access essential services, such as schools and healthcare facilities.
- **Increased security:** Installation of street lights in Ndumanu will reduce the risk of crime and enhance the overall sense of security for the residents. They will move freely during the evening, promoting a more vibrant and engaged community life. The residents of Ndumanu will be able to operate their businesses for longer hours and attract investments fostering economic growth.
- **Improved solid waste management:** A systematic approach to waste collection and transportation will ensure that residents have access to regular disposal services, minimizing the risk of uncontrolled dumping and its associated health hazards. In addition, an efficient waste management system can also generate job opportunities.
- **Improved storm water or surface run-off management**: Constructing storm drains helps channel rainwater away efficiently, preventing heavy surface run-off and reducing the risks of waterborne diseases.
- **Improved clean water supply:** Upgrading the Kihore Water Project supply facility will ensure the provision of clean and safe water to residents at affordable prices. Reduced distance and effort required for water collection, alleviates the burden on

residents, especially women and children. In addition, improved water supply technologies contribute to improved public health, minimizing the risk of waterborne diseases as well as creating employment whereby local entrepreneurs manage these upgraded water vending points.

- **Improved overall visual appeal and living conditions**: basic infrastructural improvement and maintenance for Ndumanu will foster community pride, well-being, and social cohesion. Improved visual aesthetics contribute to creating a more livable and dignified environment for the residents.

c) **Improved housing:** Land ownership documents might encourage property owners to invest in developing or improving the condition of their homes. This will enhance better living conditions and higher quality of life.

5.4.6. Potential Negative Impacts

a) **Loss of structures: Fully or partially.** The land planning and surveying process in Ndumanu may unintentionally cause disruption to people's structures. During the site visit it was observed that some structures like fences had encroached on road reserves and would require relocation.

b) **Land-related disputes:** The legal processes involved in tenure regularization can be complex and may lead to disputes and challenges. Ambiguities in land titles and property rights may persist, causing confusion and conflicts among residents.

c) **Exclusion of vulnerable groups:** The regularization process may not adequately consider the needs and rights of vulnerable groups, such as elderly, the sick, disabled, ethnic minorities, or low-income families. This can result in their exclusion exacerbating existing inequalities.

d) **Inequitable land distribution:** The regularization process might not be conducted fairly, leading to unequal distribution of land rights. The upper class in the settlement may exploit the process to secure larger and more valuable plots, leaving marginalized or vulnerable residents without land.

e) **Air Pollution:** Emissions to the atmosphere during construction and operation could affect sensitive ares which include the dispensary, the ECDE Center, and residential areas. Construction activities such as utility diversions, road excavation, and road resurfacing will produce dust and particulate emissions, which may be worsened by winds and dry conditions. Dust emissions have the potential to cause transient severe negative impacts, particularly on road users and sensitive receptors near construction sites and compounds.

f) **Noise and vibrations:** Noise and vibration can be a source of disturbance at sensitive receptors. Residential areas and places of worship need to be safeguarded.

g) **Water Pollution:** Construction activities such as the diversion of utilities, road excavation, and road widening have the potential to create pathways for pollutants to enter watercourses and indirectly impact water quality. Soil compaction during construction has the potential to increase the rate of surface water runoff.

h) **Waste Generation:** This is both solid and liquid waste generated during the construction and operation stages of the infrastructure improvement phase of the project.

i) **Soil Pollution:** Construction activities will require the excavation of existing ground and the existing roadbed, which may result generation and loss of topsoil and solid waste generation

5.5. Mitigation Measures of the Predicted Impacts

This section provides an overview of practical and economically viable strategies designed to mitigate significant adverse environmental and social impacts. Each strategy is designed to address specific challenges associated with particular situations, ensuring that the unique characteristics of each context are taken into account.

To support the implementation of these strategies, relevant legal frameworks and regulatory guidelines are referenced, including but not limited to the Environmental and Social Management Framework (ESMF), the Resettlement Policy Framework (RPF), and the Site Management Plan (SMP). These documents serve as essential tools for guiding the development and execution of the proposed strategies, ensuring compliance with established standards and best practices.

Environmental and Social Risks/Community concerns	Risk Rating	Impact Rating	Potential interventions	Applicable Legal Instruments, and WB Safeguard Policy
Loss of structures: Fully or partially	Low	Low	➢ Adopting negotiated planning standards which minimizes the number of Project Affected Persons (PAPs). ➢ Provision of facilitation through non-monetary compensation such as in-kind replacement of the structures.	➢ The World Bank Safeguard Policy on Resettlement (OP 4.12) which provides clear directives regarding compensation and resettlement with objectives aimed at minimizing the impact on affected communities. ➢ KISIP Resettlement Policy Framework which establishes guidance for addressing potential displacement resulting from project activities.
Inequitable land distribution	Low	Low	Conduct awareness campaigns to foster gender equality in property ownership.	The Land Registration Act, 2012 The Land Act, 2012 The National Lands Commission Act, 2012 The Environment and Land's Court Act, 2011
Exclusion of vulnerable groups	Low	Low	➢ Regular Community Meetings and Feedback Mechanisms to prevent marginalization and ensure equitable distribution of project benefits ➢ Implement targeted social inclusion programs and policies to address the specific needs of these groups. ➢ Provide opportunities for their input and representation in decision-making forums.	The Persons with Disabilities Act, 2003 National Social Protection Policy Framework, 2011 Involuntary Resettlement (OP 4.12) Social Assessment and Management (OP/BP 4.01)

Environmental and Social Risks/Community concerns	Risk Rating	Impact Rating	Potential interventions	Applicable Legal Instruments, and WB Safeguard Policy
			➢ Conduct awareness campaigns to foster social cohesion. ➢ Linkage of the vulnerable PAPs to existing social programs of the government including social programs .	
Land-related disputes	Low	Low	➢ Engaging Grievance Redress Committee (GRC) in resolving disputes. ➢ Ensuring all stakeholders have participated actively in planning and surveying processes.	The Land Registration Act, 2012 The Land Act, 2012 The National Lands Commission Act, 2012 The Environment and Land's Court Act, 2011
Air Pollution	Low	Low	Compliance to NEMA air quality regulations and proper use of Personal Protective Equipment (PPEs)	Environmental Management and Co-ordination (Air Quality) Regulations, 2013 OP 4.01 Category B; Public Health Act, Cap 242 (Revised Edition 2012)
Noise and vibrations	Low	Low	Compliance to NEMA noise regulations and proper use of Personal Protective Equipment (PPEs)	Environmental Management and Coordination (Noise and Excessive Vibration Pollution) (Control) Regulations, 2009 Public Health Act, Cap 242 (Revised Edition 2012)

Environmental and Social Risks/Community concerns	Risk Rating	Impact Rating	Potential interventions	Applicable Legal Instruments, and WB Safeguard Policy
Waste Generation	Low	Low	Application of Integrated solid waste management system	Environmental Management and Coordination (Waste Management) Regulations, 2006 (Legal Notice No. 121 of 2006) EMCA- Environmental (Impact Assessment and Audit) Regulations, 2003 OP 4.01 Category B; Sustainable Waste Management Act, 2022;
Water Pollution	Low	Low	Implement sustainable designs that adopt natural drainage patterns.	The Water Act, 2016, Cap 372 Public Health Act, Cap 242 (Revised Edition 2012)

Table 8: Mitigation Measures for the Predicted Impacts

CHAPTER SIX: ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN

6.1. Overview

This section outlines the Environmental and Social Management Plan (ESMP) for the Ndumanu informal settlement. The Environment and Social Management Plan is a safeguard instruments that identified the impacts resulting from the tenure regularization activities and provide the mitigation measures. The plan recommends the type of safeguard instruments to be prepared either the Social Management Plan (SMP), Resettlement Action Plan (RAP) OR Abbreviated Resettlement Action Plan (ARAP). The social management plan is prepared to mitigate the low magnitude impacts and is implemented socially through community assistance at no cost.

6.2. Objectives of the ESMP

The main objective of the ESMP is to provide mitigation measures for the identified impacts.

Specific Objectives

To identify actual impacts resulting from the tenure regularization activities
To recommend the environmental and social safeguard instruments to be prepared
To provide the mitigation measures for the identified impacts
To provide implementation and monitoring framework for the proposed mitigation measures.

6.3. Methodology of ESMP

The methodology adopted during preparation of this report was guided by the project's Resettlement Policy Framework (RPF), 2019, the World Bank's policy on Involuntary Resettlement (OP 4.12), KISIP2 Social Management Plan, 2019, EMCA 1999, Environmental and Social Management Framework (ESMF), 2019 among other relevant policies and legal frameworks.

The assessment further adopted finding and environment and Social Screening assessments prepared for the settlement where activities listed below were undertaken;

Settlement field survey for assessing environment and social risks including transect walks and observation of the settlement;

Review of the base map prepared for the settlement for various issues including incompatible land uses,

Overlays of road proposals (conventional and adoptive) on the base maps to highlight risk of displacement;

Review of the socio-economic baseline assessment/report prepared for the settlement and Consultations with relevant stakeholders including holding public meeting.

Table six presents the predicted impacts of the project, the proposed mitigation measures, the monitoring indicators, the frequency of monitoring and the institutions incharge of implementing the ESMP.

6.4. Institutional Arrangements For EMP Implementation

Table 9 shows the ESMP for the Ndumanu Informal Settlement. The key actors responsible for overseeing the implementation of the ESMP are expected to execute their roles

throughout the project period. Their role is crucial in ensuring that the plan is effectively put into action, and that environmental and social considerations are properly integrated into the project. The consultant will provide expertise, guidance, and oversight to ensure compliance with regulations, best practices, and stakeholder expectations. They will also help identify and mitigate any potential risks or issues that may arise during implementation.

Impact	Mitigation Measure	Monitoring Indicators	Monitoring Frequency	Responsibility
Environmental Negative Impacts				
Air pollution	Adherence to NEMA Air Quality Regulations and Appropriate Use of Personal Protective Equipment (PPE)	Good air quality	During infrastructure upgrade phase	KISIP CPCT, Consultant, Contractor.
Noise pollution and vibrations	Compliance to NEMA noise levels regulations Adherence to NEMA Noise Level Regulations	Recommended low noise levels not exceeding Maximum Permissible Daytime Noise Level of 60 dBA	During infrastructure upgrade phase	KISIP, Contractor.
Generation of debris	Implementation of a comprehensive solid waste management system in the settlement	Clean environment	During infrastructure upgrade phase	KISIP CPCT, Consultant, Contractor, Community members.
Social Negative Impacts				
Land related disputes	Utilize the Grievance Redress Committee (GRC) for peaceful dispute resolution. Promote transparent and inclusive community planning.	Enhanced collaboration among stakeholders. Reduced or eliminated escalation of land disputes.	During the project period	KISIP CPCT, SEC, GRC
Possible displacements	Develop settlement management plan that ensures those who may be affected willingly accept and comply with the plan proposal.	Low or no reported cases of involuntary displacement. Successful implementation of resettlement plans with community buy-in.	During the project period	KISIP2 CPCT, Consultant, SEC, GRC

Exclusion of the vulnerable group	Promote inclusivity in planning and surveying by actively engaging vulnerable groups. Develop and implement targeted social inclusion programs and policies to meet the specific needs of vulnerable groups. Guarantee their involvement in decision-making processes.	Increased participation and representation of vulnerable groups, more equitable outcomes. Increased participation of vulnerable groups in project activities. Reduced instances of marginalization or exclusion reported.	During the project period	KISIP CPCT, Consultant, SEC, GRC

Table 9: Environmental and Social Management Plan

CHAPTER SEVEN: CONCLUSIONS AND RECOMMENDATIONS

The findings from the screening process carried out in the Ndumanu informal settlement have revealed the relationship between the proposed land planning and surveying activities and their potential impacts. Positive impacts were projected to be an increase in tenure security, which would empower residents by granting them legal acknowledgment of their land rights, improved living conditions, and the subsequent upgrading of essential infrastructure, among other benefits.

The screening process has also uncovered various potential adverse impacts linked to these activities. Notably, the risk of losing existing structures poses a significant concern, as it could lead to the displacement of residents and the disruption of community bonds and land-related conflicts among others. The screening process concluded that an Environmental Impact Assessment (EIA) Study is not required at this initial stage of land planning and surveying. This conclusion is based on the assessment that the immediate impacts of these activities are manageable and do not necessitate the comprehensive analysis.

Recommendation

The subsequent sub-component of infrastructure upgrading, such as roads, sewer lines, and storm drains, which have significant implications for the environment. This report therefore recommends that the sub-componet undergo an Environmental Impact Assessment (EIA). This requirement arises because this sub-components is categorized under the Second Schedule of the Environmental Management and Coordination Act (EMCA) of 1999, as amended in 2015, specifically Legal Notices No. 31 and 32, and classified as Low Risk Projects. These sub-projects have the potential to generate adverse environmental and social effects that are confined to the project area.

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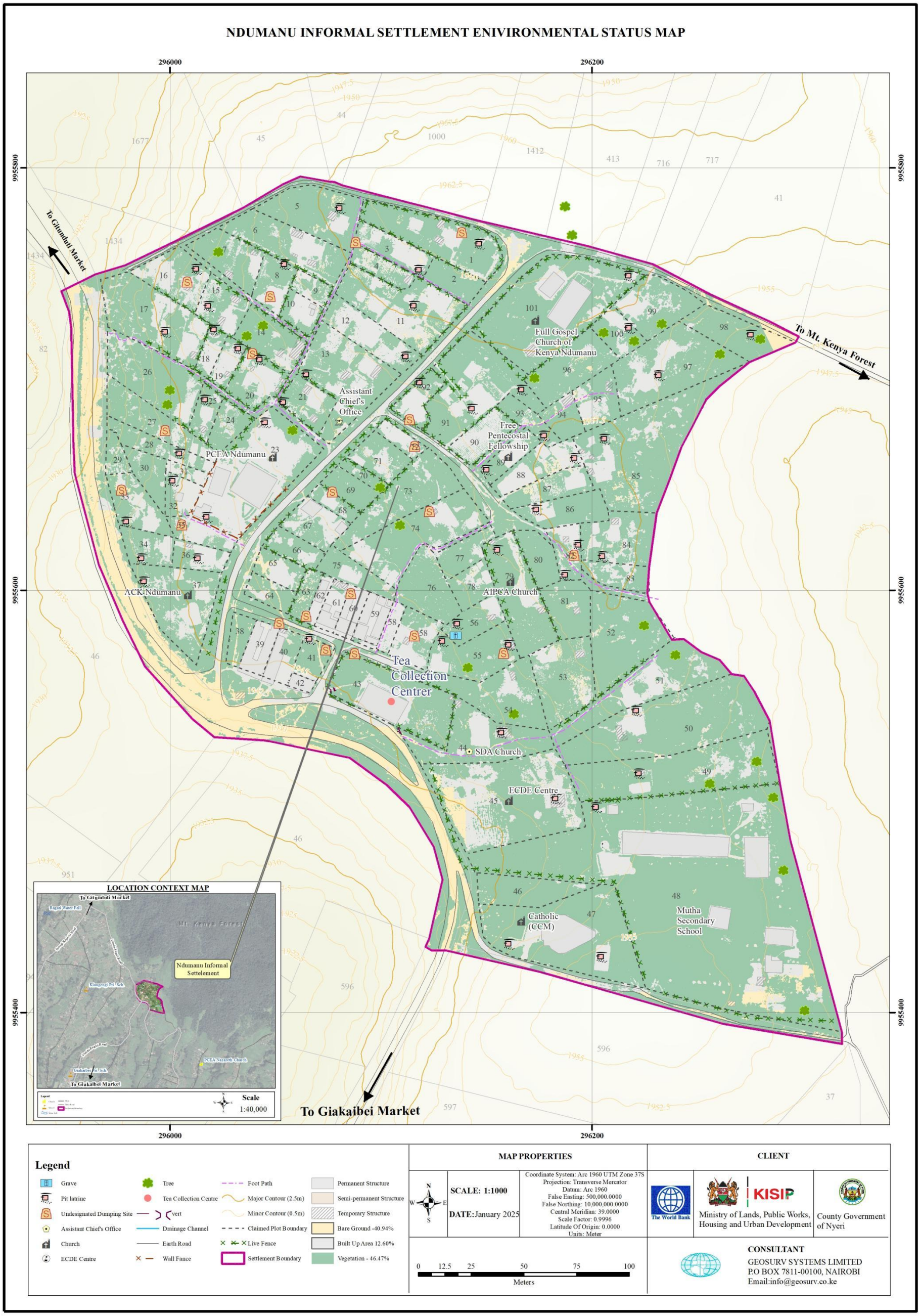
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ANNEXES

Annex 1 Environmental Status Map



Annex 2: Environmental and Social Risk Screening Checklist

Project details		
Name of the settlement	Ndumanu Informal Settlement	
Date	26th-27th November 2024	
Issues to be considered	***Yes/No. Briefly Describe***	***Is this likely to result in a significant effect?Yes/No/? -why***
A: Triggers to EMCA 1999 (Amended 2015)		
1. Does the project fall under the second schedule of EMCA Cap. 387	Yes. Planning and surveying will not require EIA however subsequent infrastructure upgrading phase will require EIA. The second schedule of EMCA legal notice 31 and 32 stipulates that every project must be assessed for environmental and social impacts. Based on the significance of impact rating, the project is classified as medium risk.	No. Environmental and social impacts expected to arise from physical planning and surveying are not significant hence, does not require EIA. Although, infrastructure sub projects are classified as medium risk as prescribed on Second Schedule of EMCA 2015, Legal Notice No. 31 these will require EIA. Further, Legal Notice No. 31 section 7(1) and (4) dictate that a comprehensive impact assessment should be carried out.
B: Triggers to World Bank Safeguard Policies		
Does the project trigger one or more ofthe WB Safeguard policies	Yes, OP 4.01: Environmental Assessment and OP 4.12: Involuntary Resettlement	No, the project is classified as category B which are less impacts on the environment and are site specific. Displacement might be minimal to none by adopting a proactive approach to planning that will help reduce displacement and relocation issues.

C: Sectoral Policies and Laws applicable		
Does the project fall under/trigger anyother GOK Policies and Laws?	Yes, • Vision 2030 • Sessional Paper No. 3 of 2009 on National Land Policy • Sessional Paper No. 3 on National Housing Policy for Kenya • The Constitution of Kenya (Articles 42 and 70) • The Environment Management and Co-ordination (Amendment) Act, 2015 and its Subsidiary Regulations • Environmental Management and Coordination (Water Quality) Regulations, 2006 • Environmental Management and Coordination Act (Environmental Impact Assessment and Audit) (Amendment) Regulations of 2019 • Physical and Land Use Planning Act, 2019 • The Occupational Safety and • Healthy Act, No. 15 of 2007 • Environmental Management and	Yes, these policies and laws provide a framework for evaluating potential impacts and implementing necessary mitigation measures.

	Coordination (Air Quality) Regulations, 2014 • Environmental Management and Coordination (Noise and Excessive Vibration Pollution Control (Regulations) 2009 • County Government Act, 2012 • Public Health Act Cap 242 • The Water Act, 2016 • HIV and AIDS Prevention and Control Act 2006 (Revised 2012) • The Urban Areas and Cities (Amendment) Act 2019 • Public Participation Act, 2018 • Sexual Offences Act, 2006 • Labour Relations Act, 2007 (Revised) 2012 • Physical and Land Use Planning Act,	
D: Project Location		
4. Is the proposed site a protected or reserved site (Provide proximity in kms) • Biosphere Reserve	Yes. Mount Kenya Forest approximately 1 kilometers away.	No. Anticipated impacts will not affect the forest, the forest is also protected.

• National Park • Wildlife/Bird Sanctuary • Wetland • Important Bird Area • Coastal area with corals • Mangrove areas (or estuary with mangroves) • Natural Lakes • Habitat of Migratory birds (outside protected area) • Migratory route of wild animals • Area with threatened or rare/endangered flora (outside protected area) • Area with threatened or rare/endangered fauna (outside protected area) • Reserved / Protected Forest • Zoological Park/Botanical		

Garden		
Are there any areas on or around the location which are protected under international or national or local legislation for their ecological, landscape, cultural or other value, which could be affected by the project?	Yes. Mount Kenya Forest .	No
Are there any areas on or around the location which are important or sensitive for reasons of their ecology e.g. wetlands, watercourses, coastal zone, mountains, mangroves, forests or woodlands, migratory routes which could be affected by the project?	No	No
Are there any routes or facilities on or around the location which are used by the public for access to recreation or other facilities, which could be affected by the project?	Yes, roads that provide access to facilities such as the Tea Buying Center,the chiefs office, Muthea High School and the places of worship will be affected by the project.	Yes. Access to the facilities will be improved for the better of the community.
Is the project in a location where it is likely to be highly visible to many people?	Yes, it is located off the Itundu-Ragati Road therefore easily accessed by many people.	Yes, tenure regularization will enable property owners develop their land with modern structures thus improving the overall aesthetics of the settlement.

Are there any areas or features of historical or cultural importance on or around the location which could be affected by the project?	No	No
Is the project located in a previously underdeveloped area where there will be loss of green field land?	No, the area has been a residential and agricultural site.	
Are there existing land uses on or around the location e.g., homes, gardens, private property, industry, commerce, recreation, public open space, community facilities, agriculture, forestry, tourism, mining, or quarrying which could be affected by the project?	Yes, there are residential, commercial, educational, religious, public purpose and agricultural land uses.	Yes, project interventions will improve the overall status of the settlement, for example, land rates will increase, aesthetics of the settlement will improve.
Are there any Plans for future land uses on or around the location which could be affected by the project?	No	
Are there any areas on or around the location which are densely populated orbuilt up, which could be affected by the project?	No. The population density is evenly distributed in the area.	Yes, tenure regularization might displace some people and structures in order to improve the quality of life.
Are there any areas on or around the location which are occupied by	Yes, sensitive institutions include, Muthea High School,	Yes, during construction there will be noise and air pollution, on the other hand, there will be improved

sensitive land uses e.g., hospitals, schools, places of worship, community facilities which could be affected by the project?	St. Joseph's Catholic Church, Chief's Office, SDA Church, Faith Manifestation Church, PCEA Church and Ndumanu Tea Buying Center among others.	access, and security for these facilities.
. Are there areas on or around the location which are subject to pollution or environmental damage e.g., where existing legal environmental standards are exceeded, which could be affected by the project?	No	
. Is the site already degraded (low groundwater, poor soil quality)?	No	
. Are there steep slopes in the proximity of the investment site?	Yes. They are only used for agricultural land uses.	No. The steep slopes are only used for agricultural hence no risk to human settlement.
. Do people live on the proposed site?	Yes, the settlement has residential land use.	Yes, during construction effects might include noise, dust, blocked access to premises. During construction there will be an increase in the number of residents as well as increased economic growth.
. Do indigenous peoples live on or near the site?	Yes, the indigenous people living on the site are the Kikuyu tribe	
. Is the site vulnerable to natural hazards (in floodplain, near volcano, on seismic fault, near	No	

coastline in hurricane zone)?		
. Are there land title conflicts?	No, the settlement is on uncontested public land	
. Are there known archaeological, historical or other cultural property? Areany of these world heritage/ UNESCO designated, etc.?	No	
E: Construction impacts		
. Will construction, operation or decommissioning of the project involve actions which will cause physical changes in the locality (topography, land use, changes in waterbodies, etc.)?	No,	
. Will the construction or operation of the project use natural resources such as land, water, materials or energy, especially any resources which are non-renewable or in short supply?	Yes, construction will use, water, sand, land and timber.	Yes. Extraction of these resources will have significant environmental impacts over time.
. Will the project involve use, storage, transport, handling or production of substances or materials which could be harmful to human health or the	Yes, hazardous chemicals, such as oil, lubricants, adhesives, paints and coatings will be used.	Yes. May cause soil pollution Health risks such as respiratory diseases

environment or raise concerns about actual or perceived risks to human health?		
. Will the project produce solid wastes during construction or operation or decommissioning?	Yes, significant amount of waste will be generated	Yes. Lack of waste facilities in the settlement may contribute continued dumping of solid waste
. Will the project release pollutants or any other hazardous, toxic or noxious substances to the air?	Yes, use of machinery will produce greenhouse gases	Yes. Air pollution Health risks like respiratory diseases
. Will the project cause noise and vibration or release of light, heat energy or electromagnetic energy?	Yes, especially when the works are near schools, places of worship and residential areas.	Yes. Noise pollution
. Will the project lead to risks of contamination of land or water from releases of pollutants onto the ground or into surface waters, ground water, coastal waters or the sea?	Yes, infiltration during road construction.	Yes. Water pollution is a risk
. Will there be any risks of accidents during the construction or operation of the project which could affect human health and the environment?	Yes, work place related injuries and contamination accidents may occur when construction is ongoing	No. The significance will not be much because the applicable laws e.g. OSH Act will safeguard construction workers. Not significant but the contaminants may infiltrate in to the sub soil

. Will the project result in social changes, for example, in demography, traditional lifestyles, employment?	Yes, population influx from other area to reside in the settlement. The project will create employment for the youth for instance	Yes, tenure regularization will lead to increased population, property rights improves that status of the land owner hence change in lifestyle. During the planned infrastructure improvement employment will be created.
F: Water Resource Impacts		
. Could the investment result in a modification of groundwater levels by altering flows, paving surfaces or increasing water extraction?	No	
33. Could it affect groundwater quality?	No	
. Could it affect quality (through sediment, wastewater, storm discharge or solid waste) of nearby surface waters (lake, rivers, streams)?	No	
. Will it affect water quantity in nearbywater bodies (lake, river, stream)?	No	
. Are there nearby potable water sources that need to be protected?	No	

G: Drainage impacts		
. Will the investment in storm water drainage affect existing drainage patterns?	No	
. Will it cause standing water, which couldcause public health risks?	No	
. Will erosion result in sediment discharge to nearby water bodies?	No	
. Will surface drainage patterns be affected in borrow pits and quarries?	No	
41. Will infiltration patterns be affected?	No	
H: Ecosystem impacts		
. Could the investment affect natural habitats or areas of high ecological value?	No	
. Could it affect natural characteristics of adjacent or nearby sites?	No	
. Could it affect wildlife or naturalvegetation?	No	
I: Socio-economic impacts		

. Will the project entail resettlement of population?	No	
. Will the project affect People's property or livelihoods/income?	To be determined after planning and surveying of the settlement. However, the project will create employment opportunities for the local people.	If encroachment is minimal displacement will be significant
. Will the project affect indigenous peoples?	No	
. Will it limit access to natural resources to local populations?	No	
49. Will it have an impact on land use?	No	
. Will it induce further encroachment of nearby areas?	No	
51. Will it cause any health impacts?	No	
. Will it disturb nearby communities duringconstruction?	No,	
53. Could cultural resources be affected?	No	
54. Could it affect nearby properties?	No	
J: Operation impacts		

. Is the project susceptible to earthquakes, subsidence, landslides, erosion, flooding and extreme or adverse climatic conditions e.g. temperature inversions, fogs, severe winds, which could cause the project to present environmental problems?	No	
. Are there any other factors which should be considered such as consequential development which could lead to environmental effects or the potential for cumulative impacts with other existing or planned activities in the locality?	No	
K: Displacement impacts		
57. Acquisition of private/community land?	To be determined during planning	
. Alienation of any type of government land including that owned by urban localbody?	To be determined during planning	
59. Clearance of encroachment from government/ urban local body land?	To be determined during planning	
. Clearance of squatting from Government/Urban	To be determined during planning	

local body?		
. Number of structures, both authorized and/or unauthorized to be acquired/cleared?	To be determined during planning	
62. Number of households to be displaced?	To be determined during planning	
. Details of village common properties to be alienated, Pasture land (acres) cremation /burial ground and others specify?	To be determined during planning	
64. Describe existing land uses on and around the project area (e.g. Community facilities, agriculture, tourism, private property)?	Agricultural Commercial Residential Educational Public Purpose	
. Will the project result in construction of workers or other people moving into or having access to the area (for a long period and in large numbers compared to permanent residents)?	No, construction workers will be at the settlement during the construction phase	
. Are financial/in kind compensation measures	To be determined during planning	

expected to be needed?		
L: Loss of Assets, Crops, fruit, household infrastructure and livelihood		
. Will the project result in the permanent or temporary loss of.		
. Crops,	No	
. Fruit trees/coconut palms? Specify with numbers	No	
. Households assets/infrastructure? Specify with numbers	No	
. Loss of agriculture land? specify with numbers.	No	
M: Public and Occupational health and safety, welfare, employment and gender		
. Is the project likely to provide local employment opportunities, including employment opportunities for women?	Yes. Planning and surveying will create temporary job opportunities for local youths and women particularly during social economic survey and installation of beacons. Other new job opportunities will also emerge during infrastructure upgrade. The project will for promote gender equality and empower women economically	Improved living standards for the women.
. Is the project being planned with sufficient attention to local poverty alleviation	Yes, the project has a socio-economic inclusivity component that aims to address	Improved status of living for the residents

objectives?	poverty levels in the settlement/	
. Is the project being designed with sufficient local participation of women in the planning design and implementation process?	Yes,Women are actively involved in the project, especially in roles within SEC and GRC. They will be further involved in household socio-economic surveys, focused group discussions, and all public participation forums.	Women's participation in project will be crucial for promoting gender equality and harnessing the diverse perspectives and talents of all individuals
. Will the project affect/lead to traffic and Pedestrian Safety?	Yes. During preparation Local Physical and Land Use Development Plan for the settlement, pedestrian safety will be considered by providing walk ways.	
. Will the project Interfere with the normal health and safety of the worker/employee/public?	No. The project will ensure appropriate use of Personal Protective Equipment (PPEs), proper signage, provide insurance cover and on site first aid kits for its workers.	
. Will the project introduce new practices and habits?	Yes, solid waste disposal and handling methods	Improved solid waste management practices
. Will the project lead to child delinquency(school drop-outs, child abuse, child labour, etc.?	No	No
79. Will the project lead to gender disparity?	No, the project is taking steps to ensure women rights are championed in tenure regularization and the process is all inclusive	Increased gender equality in property ownership

. Will the project lead to social ills (drug abuse, excessive alcohol, consumption,crime, etc.)?	No	No
N: Historical, Archaeological, or cultural Heritage sites		
. Based on available sources, consultation with local Authorities, local knowledge and/ or observation could the project alter?	No	No
82. Historical heritage site(s) or require excavation near the same?	No	No
. Archaeological heritage site(s) or requireexcavation near the same?	No	No
. Cultural heritage site(s) or require excavation near the same.	No	No
. Graves or sacred locations or require excavation near the same?	No	No
O: Result/Outcome of Environmental and Social Screening Exercise		
Environmental Impact Assessment Required or not	✓ No Environmental Impact Assessment Required for land planning and surveying ✓ Infrastructure sub projects are classified as low risk and are prescribed on Second	Environmental and social impacts expected to arise from physical planning and surveying are not significant hence, does not require EIA.

	Schedule of EMCA 2015, Legal Notice No. 31 these will require EIA. Further, Legal Notice No. 31 section 7(1) and (4) dictate that a comprehensive impact assessment should be carried out.	
Social Management Plan Required	Yes. To mitigate potential social impacts.	
RAP category required (RAP/ARAP)	To be determined during the planning stage	
Vulnerable and Marginalized Groups Plan required	To be determined during the planning stage	
Any special conditions.		
P: Authorisation		
Screening undertaken by: Designation:		Signature: Date:
Approved by: _____ Designation: _____		Signature: Date:
PMU confirmation by: _____ Designation: _____		Signature: Date:

Summary of features of project and its location indicating the need for EIA